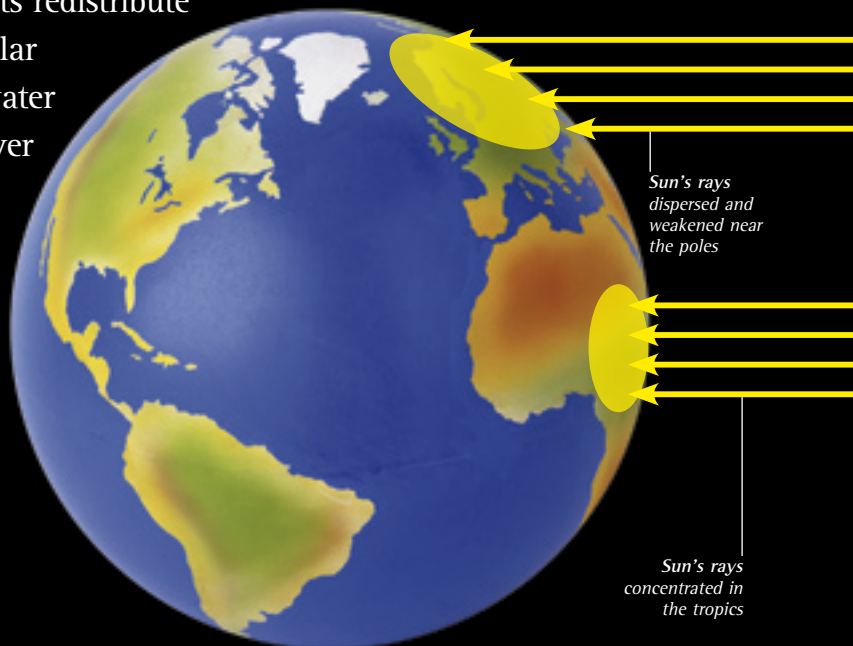


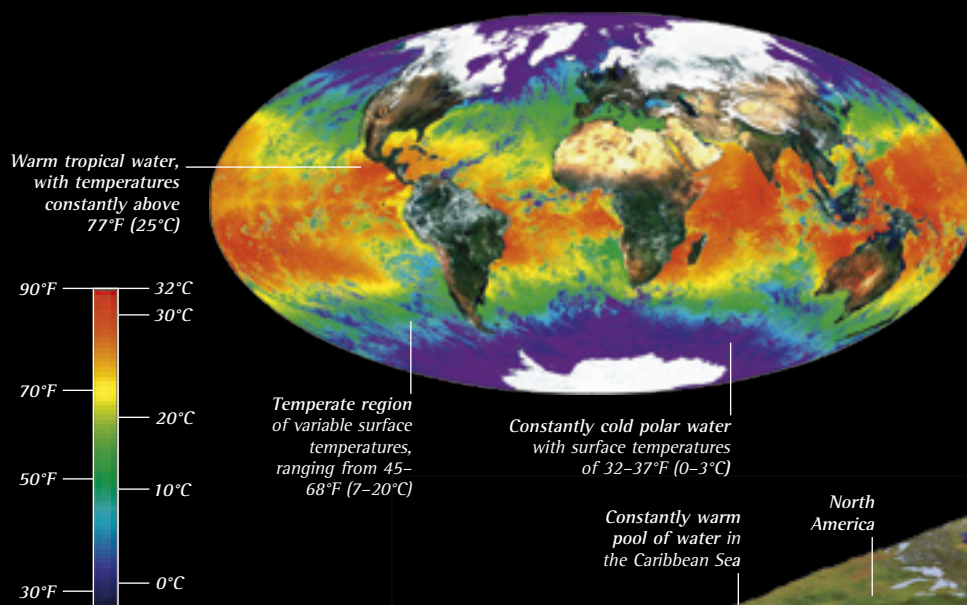
HEAT AND LIGHT

The oceans are warmed by the Sun, but tropical oceans are warmed far more than polar seas. Ocean currents redistribute this heat around the globe, which makes the polar regions warmer and the tropics cooler. Ocean water also warms up and cools down slowly, so it never gets as hot or cold as the land. Even sea ice is warmer than the ice formed on land. However, heat and light do not penetrate into the ocean depths, which are permanently cold and dark.



HEATING THE OCEANS ►

The Sun shines down directly on the tropics, warming the surface water of tropical oceans to temperatures of 86°F (30°C) or more. Closer to the poles, sunlight is spread out over a broader area, weakening its power even in summer. The polar oceans get so little sunlight in winter that they freeze over, creating vast expanses of sea ice.



◄ OCEAN TEMPERATURES

The surface water of oceans is warmest in the tropics and coldest near the poles. But the water in the ocean depths is always just above freezing point, even near the equator. Despite this, the total range of ocean temperature is only about 104°F (40°C), compared to the temperature range of 295°F (146°C) on the continents.

TEMPERATURE AND LAYERS ►

As ocean water is warmed by the Sun, it expands and becomes less dense. This makes it weigh less, so it floats on top of colder, denser water. The warm surface water is divided from deeper, cooler water by a boundary called the thermocline, which exists all year round in the tropics. It stops deep water from mixing with surface water. As a result, plant nutrients cannot reach the sunlit surface to fuel the growth of microscopic plankton. This is why tropical oceans are so clear. In colder oceans, the thermocline breaks down in the fall, allowing nutrients to the surface where they support far more life.

